

ActInf Textbook Group ~ Cohort 1 ~ Meeting 7 (Chapter 3 pt. 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 7 (Chapter 3 pt. 2) | 1:01:25

Hello everyone, it is June 16th, 2022, and it's meeting 7 of the Active Textbook Group, Cohort 1.

We're in our second discussion of Chapter 3.

Today we'll be looking at the questions that haven't been addressed, so the ones without thumbs, and also it's not too late to add more questions or to upvote questions.

And then additionally, since the Math Learning Group has been quite active in providing synchronous affordances for conversation around some of the background on the math, as well as some aspects of the equations that we'll go through today, and also even to applications of the math and developing notebooks and such, we'll take a look at the math in our move through Chapter 3.

Before we go to the questions, does anyone have anything that they just want to remark on Chapter 3?

Ali, and then anyone else?

Yeah, well, last week I talked about how I changed my mind about these bottom-up and top-down characterizations of low road and high road respectively after reading Chapter 3, but it turns out that they are in fact bottom-up and top-down approaches after all.

You see, I came across this, in my opinion, undeservedly lesser-known piece of beautiful, beautiful writing by Carl Fristen, which explicitly argues for such characterizations and the philosophical importance of this distinction.

It's called Beyond the Desert Landscape, and it's inconspicuously hidden in this book, Andy Clark and his critics.

So, I'd like to quote a couple of sentences from this writing.

Well, he writes, quote,

The high road stands in for a top-down approach that starts by asking fundamental questions about the necessary properties things must possess if they exist.

Using mathematical variational principles, one can then show that existence is an embodied exchange of a creature with its environment.

That necessarily entails predictive processing as one aspect of a self-evidencing mechanics.

The low road is the purpose the agenda established by, is to pursue the agenda established by Kant and Helholzson generalize in a bottom-up way the capacity for inference and prediction to see how far it takes us in understanding embodied exchange with the environment.

So, if anyone likes to gain a deeper understanding about these conceptual characterizations,

I highly, highly recommend this chapter of this book and quite unusually for Fristen writing.

It also lacks any mathematical formalism.

It doesn't include a single mathematical equation there.

Thank you.

Yeah, in a recent stream we also talked about the desert landscape and Clark's concern was that we might do away with what populates our implicitly assumed to be rich ecosystems of psychology, philosophy, and so on with important ecosystem players like goals.

And so, the fear or concern was that we might be in a Quinean desert landscape if we replace some of these, again, implicitly or explicitly stated to be rich constructs like goals and we're like, well, it's just distribution matching.

If we're not just a certain way, then we've made some sort of like a desert monoculture and that would be depauperate of what?

What does Clark say to his desert living critics?

All right.

If anyone else has a thought, they can raise their hand.

Otherwise, we can look at the questions.

Okay.

So, this first question.

If one defines preferred states as expected states, one can say living organisms must minimize the surprise of their sensory observations.

I'm not clear on the ontology here.

Is it that preference is the same as expectation or that formally we can use them in the same equation?

Is that an ontological claim?

And is the free energy and physics an analogy or an ontological assertion about how things really are?

So, one starting place was, I found the slide and so I wanted to highlight it.

This was in Free Energy, a user's guide, which was live stream number 37.

The slides will be probably a little bigger.

It's worth restating just how unusual it is to interpret P as a measure of both probabilities and preferences.

They both start with P , though.

There's nothing wrong with treating a distribution as a measure of preferences, because distributions can have different interpretations.

But what's unorthodox and in need of justification is giving the same mathematical term two different interpretations within the same equation.

They then add some more details that unpack a little bit about how it is the case that the same term P can be utilized as both a probability and a preference.

And then they even extended it into a third P , the pH fitness, where by allowing your probabilities and your preferences to converge, to reduce the divergence between those two distributions, then fitness is achieved.

And so they provided a restatement of the way that that is implemented,

and then highlighted an additional grounding or interpretive line.

So that's at least one part of that question.

And also to look towards the equations,

it's something that we'll be able to have very clear answers for,

and thanks to the work of Ali and Blue and others who have been annotating equations.

So for example, in equation 3.1, or we can look at like box one, equation one,

which is like a Markov blanket definition.

In the description field,

we have a natural language

description,

description,

and then the terms that are used

in the description

are those that are referenced using variables.

So that will help us

translate across natural language of different human languages,

the discrete and continuous time formalisms,

which we're going to explore in this textbook,

also

identical or isomorphic or related

formalisms in other papers,

which sometimes are using the same notation letters,

but not always.

But we could have another column for

first in 2019.

And here's first in 2013.

And then there's a conversation to be had about where they align or don't.

And that's the kind of annotation work that somebody can engage in from publications.

And that also helps us connect it to the programmatic implementations,

like in ActiveBlockFrance,

or in other scripts,

so that even that we can promote norms and practices around variable naming,

in math,

and in programming,

but also where there is divergences or differences,

which are important for innovating and evolving the norms,

we can also map them pretty coherently.

So all the equations for three can be read naturally.

So we discussed this a lot in the math group,
and again,
a lot of people contributed to this,
with one of the goals or functions being that whenever we are going to bring up an equation,
like if there's a question about equation 3.1,
we can just briefly read the current state of the description.
Because whether you're super familiar with the letters and variables and how they're concatenated,
or whether it's just like a first pass on your seeing this equation,
everybody will benefit when we align our regime of attention to actually what it says,
and then that will help us do many,
many other things.

But to return back to this question,
does anyone want to raise their hand and give a thought on the relationship between preference and
expectation,
ontological claims about preference,
and its relationship with expectation?

And more generally,
is the free energy
something that is an analogy
or mathematical isomorphism
or an equation that rhymes?

Or is it also an assertion?
It's an equation that rhymes.

What makes you say that?

No, no, just because
just because it's cute, that's all.

I just joined the conversation,
but preference and expectation,
I wouldn't think they were
at all referring to the same thing.

I mean, expectation is a mathematical operator.

Preference is
a statement.

Yes.

Expectation is used in multiple ways.

It can refer to
predictions
about the mean

or the mode
or other attributes
of a future time point.
Like, I'm expecting,
you know,
what are you expecting to happen
on next Monday?
Right, right, right.
Also, expectation can refer,
and that's how it's used
with the fancy E,
to refer to
the expected value
of a distribution,
which for a Gaussian
is the mode
and the center
of a symmetric peak.
and that doesn't need
to be about
a future time point.
That can just be
the expectation
of the height
of the classroom
right now.
It's not saying
that's what we're expecting
it to be
in a year
as it might be used
conversationally,
but it's referring
to just like the average
of a distribution.
Right.
Mathematically,
it's an operator,

and verbally,
it can be
a prediction
or a desire
or something.

Yes.

Yes.

And then,
yeah,
again,
in 37 and elsewhere,
it'll be worth
returning to this
again and again
because it's one
of the essential
architectural
decisions
in active inference
that enable
it's part
of the necessity
and sufficiency
complex.

I don't know
whether this is
the whole
meat of it,
but it's part
of what allows
the dispensal
of like
an explicit
reward term.

Like,
a homeostatic
reinforcement
reward-driven

learner
would be like,
I'm more
rewarded by being
in my homeostatic
range.
And then,
there needs to be
this architecture
around,
well,
how are we
going to
pursue reward?
Alternatively,
an organism
trying to maintain
homeostasis,
realism,
or us
modeling an organism
that is maintaining
homeostasis,
instrumentalism,
we can just say
it expects
to be
in the range
that conversationally
we're going
to describe
as a preference.
How do we
put the animal
in the room
and it's
spending a lot
of its time

on this
warmer side?
It prefers
to be there.
We also
expect it
to be there.
We're less
surprised when
we see it
there.
So then,
it's possible
to,
as they write
here,
interpret
that
P
location
distribution
in the room
for the animal
that we're
observing
or the
temperature
distribution
for the
body
in the
homeostatic
context
as being
a probability
density
over temperatures,
over locations,

as well as
reflecting
the preferences,
not the
qualia or the
phenomenology or
anything like
too far out
about what it
means to prefer,
where preferences
arise from,
whether they're
fixed or
learned in the
model,
all these
many,
many trails
that can come
out of that,
but just at
this essential
layer,
we don't need
to interpolate
a reward term
between what
is expected
and what is
preferred.

Any other
comments from
someone?

Yes, Eric,
and then anyone
else?

Yeah,

just to
remark on
that,
it seems
you're still
not removing
the free
variable of
what the
preference of
the agent
or organism
is.

You're just
putting it
in a different
place so
that you
can use
it in
this unified
equation,
but you
still have
this,
it's a free
variable,
okay,
what do you
want to do?
okay,
I'm going
to call
that an
expectation
and that's
what I'm
going to

try to
drive for
when I
optimize my
variational
free energy.
But in
that sense,
it's kind
of a
convenience
or it's
a trick,
it's a
little bit
of an
unnatural
act,
I don't
know,
the way I
think of
it,
for a
purpose,
but there's
nothing,
you haven't
solved any
other problem,
the deeper
problem of
what should
the organism
attempt to
achieve,
the goals.
Yes,

thanks,

Ali.

Well,

I think

it also

might be

quite helpful

to refer

back to

chapter 2

on page

36,

I think,

yeah,

page 36.

I think,

well,

here,

they have

actually

distinguished

between

the term

prediction

and,

exactly,

yes,

here,

the term

prediction

and expectation

quite explicitly

in terms

of equation

2.6,

so.

Okay.

so let's

look at
equation

2.6.

Okay.

A few of

the,

we're all,

all,

we don't

have the

third line

fully

described,

but just,

here's the

third line.

What does

this speak

to with

respect to

preferences

and

expectations?

But here,

the preference

actually is

in the

second term,

the

callback

divergence

here,

yeah.

And so,

it's actually

inherent in

the second

term,

and the
expected
ambiguity
as it
speaks for
itself
is the
first term.

we have
here,
the risk
is defined
as the
divergence
between
the preference
and the
states.

So,
I think
that we can
see the
preference
in this
equation
as well.

one note
here is
like that
this g
is equal
to the
first,
the second,
and
this line.

But one

way that
we can
understand the
relationship of
the second
and the third
line is
like,
they're very
similar.

The first
term is
identical.

Here,
the only
difference is
that instead
of a y ,
we see an
 x .

and so
this is
like x
being the
hidden state
and y
being the
observation.

In the
case where
it's a
fully
observable
Markov
process,
then the
outcomes
and the

hidden states
are going
to be
like equal
uncertainty
about them
because
in that
case,
it's
functionally
that the
hidden states
are observables.

However,
if there's
any amount
of ambiguity
like the
A matrix
between the
hidden state
and the
observable,
then there's
always going
to be more
uncertainty
about the
hidden state
relative to
the observable.

And then,
importantly,
on both
sides,
there's a
conditioning

vertical line
on policy
selection.
and so
that is
what prevents
us from
getting into
like,
well,
what if
it's 95%
correct or
96% or
97% correct
or any
of these
sort of
model
diminishing
returns
debates?
It's a
functional
that's calculated
for each
policy that
have been
possible of
composing
given the
affordances.
A policy is
just a sequence
of actions,
so if
there's
four

movement
affordances,
there's
four policies
of time
depth one,
there's 16
policies of
time depth
two,
and so on.
And for
whatever in
this discrete
formalism,
finite set of
policies there
are,
they're evaluated
here.

And then
there's a
policy
selection
that could
choose the
lowest
expected free
energy or
it could
proportionally
choose from
them with
different kinds
of weighting
and so on.
But this is
comparing the

relative expected
ambiguity and
risk
conditioned upon
different specific
policies.

And there's a
process by which
those specific
policies can be
enumerated,
and then this
is a process by
which they're
going to be
evaluated,
and then
there's a
policy selection
by which
they're
selected.

So,
organisms that
fail to do
this adequately
enough will
be dissipated,
or systems,
whether it's
just a
molecular system
or some
other kind
of statistical
system that
fail to do
this well

enough to
remain being
that kind
of thing,
are not
amenable to
repeated
measurements.

They are not
that thing.

But things
that remain
being that
thing for
the time
horizon of
observations
will be
selecting
policies that
are consistent
with this.

Can I ask a
question about
this?

Yeah.

About this
equation here?

Say, like,
the second
line.

So, I would
have expected...

So, first of
I'll see, the
preferences is not
really spelled
out very much

in this chapter
and I don't
know where it
will be.

But I would
have expected
something more
like preferences
given observations.

Like, you know,
the observations
of what's
happened in the
world that we
can measure.

That's the
observation.

Why?

And how does
that fit with
my preferences?

So, what I
want to optimize
is my preference,
the achievement
of my preferences
as delivered
by the
observations.

Now, I
understand that
in that second
equation, you're
going to do a
divergence between
observations, the
tildes, the
y-tildes.

So, you can't
flip around P of
 C given y -tilde
there.

So, can somebody
explain to me how
this all fits
together then?

How do you
achieve preferences
under that
equation?

I'll give
one thought.

And I
absolutely agree
to the first
point, which is
like, C is
just like, I
don't even know

if it's
described in
this chapter.

Like, it just
given...

34, I
think.

34?

Let me check
it again.

I mean, there
is a C .

He said,
they say

the C .

parameter
includes

preferences, but
that's all.

It includes
parameter.

Yeah.

And also, if
you go to
section 7.4, I
think, after
7.

Yes.

Yeah.

In that
section, it is
untangled why
these two
parameters are
related and how
these two
parameters are
related to each
other.

Yeah.

In section...

Yeah.

Yeah.

I mean, like,
there's...

That's like
multiple scales
of where the
information is
not provided
before when it
needs to be
provided, which
is like...

This I have

made as just a helpful diagram of the partially observable Markov decision process, which is C is the preferences, and it influences how G evaluates policies which are generated from affordances over a given time horizon. policies intervene upon how hidden states change in the world.

And there's more in different readings.

But to return to Eric's question, and then again, anyone with the raised hand.

So, the question was, whether we're comparing outcomes or states, how does this second term help us realize our preference?

Well, when is the KL...

We're trying to minimize G of

pi.

Better policies
have a lower
score.

It's like golf,

I guess.

Or you could
add negatives
and make it
higher.

And also,
just, it's an
interesting note,
like climbing
mount improbable.

Biologists often
think about the
fitness peak and
climbing to
fitness peaks,
whereas for a
physicist, it's
often framed,
and also, like,
computer science,
gradient
descending, and
local minima,
global minima,
but they're just
actually, like,
the same landscape
with some negative
signs.

So, the KL
divergence, and
here we're trying
to minimize.

If it's zero,
that would mean,
like, we are
perfectly realizing
our preference.
So, let's just
say that there's
two policies that
we're comparing
between, and one
of them is going
to perfectly
realize our
preferences, the
other one isn't.
so that we have
like π_1 and
 π_2 .
And we're going
to be calculating
the KL
divergence of the
outcomes that we're
expecting under the
variational distribution
queue under policy
1 and under
policy 2.
So, if policy
1 is going to
perfectly realize
our, if the
outcomes of policy
1 are expected
to perfectly realize
our preferences,
this term will
be zero.

Whereas, if
policy 2 is
going to be
expected to
result in an
outcome distribution
that's not
aligned with our
preferences, it
doesn't get
penalized here.

This is just
evaluated as a
value.

Nor does it get
penalized here.

Because in this
case of fixed
preferences, that's
also not changing
as we iterate
through policies.

However, there
would be a
divergence that
would be non-zero,
like all
divergences are,
between the
inferior policy
and the
superior policy.

So, by
iterating over
the finite set
of policies that
we have,
taking this

term and just
leaving it to the
side for now,
only the KL
divergence, this
will lead us to
select policies
that can be
expected to
bring alignment
between the
observations
conditioned on
policy and
our preferences,
preferred outcome
distribution.

Thank you.

That's very
helpful.

So, just to
kind of say that
slightly differently
is, given your
preferences, that's
going to imply
some outcomes.

And what you're
trying to do with
that KL
divergence is you're
trying to get the
policy that will
achieve those
outcomes.

things.

So, if we
expect and

prefer to be within the homeostatic thermal range, and then the two policies in the snowstorm are add the jacket or remove the jacket, which one of those are expected to have the minimum divergence between us being in the homeostatic range, and then the policy-conditioned expected outcomes, I'm going to have less divergence. It doesn't even mean you're going to live. It just means I'm going to have less divergence by having the jacket on in the snowstorm than by taking it off, which is going to move the expected outcome distribution further from the preference distribution, which here is fixed, but doesn't have to be fixed.

Okay.

So, we have
several.

Is anyone here, the
person who has
written any of these
questions, and then
we can just go to
that one?

Lyle?

Yes.

There we go.

I think I'm
going here.

Okay.

So, yeah, I
wrote the one
that is referring
to the living
organisms and the
question about to
what extent are
these techniques
valuable in what
I would call social
systems of
wide-ranging
sorts.

And I know that
this book at the
outset limits
itself to
biological systems.

At least that's my
recollection from
the introduction.

But it just seems
obvious to me that

this would have much
broader applicability.

I'm curious what
sort of what the
state of that is.

do I hesitate to put
anyone on the spot?

John, would you
give a first
response there while
I add some
resources?

Yeah.

My first thought
on that is
obviously our
civilization is
struggling.
civilizations have
failed in the past
to collapse.

So it's not
necessarily the
case that every
social structure or
societal structure is
actually performing
functional active
inference.

inference.

But if we were a
highly functional
organism, if
societies were a
highly functional
organism, we would
be doing something
like active

inference and
learning from
our mistakes and
making sure that
we put ourselves
in an environment
that is sustainable
and all that kind
of stuff.

So my first
thought is I
think that all
of this is
applicable to
societal systems,
but it doesn't
necessarily mean
that any existing
societal systems
actually perform
functional active
inference.

Yeah.

Thanks.

It's almost like
saying, okay,
we're studying the
penguins and their
thermostats and
homeostats.

This model could
be parameterized to
fit a thriving or
a failing society
along various
different dimensions.

And so in the
sense that this is

like a linear regression that can capture failing as well as succeeding societies, yes, the nested Markov blanket formalism has qualitatively but increasingly quantitatively been applied and there's multiple like startups and applications that are explicitly pursuing this angle.

It's been brought up for several years now and in 2020, several of us wrote about it being applied to remote teams and organizations and communities.

I can add these links to decentralized science as a system and ecosystem.

So, yes, people have started to sketch that continent out and there's a lot more to understand.

What do you think,
Lyle?

Yeah, I think
that makes a ton of
sense.

I think, obviously,
I guess the way I
was thinking about
the question is not
so much whether to
what extent social
systems do this
because social
systems, so you
could take two
points of view.

One is you could
say social systems
are behaving
cognitively in the
ways that we're
discussing.

In other words,
there are
preferred states
that are driving
behavior.

Now, we could
say those
preferred states
are the wrong
states for
whatever reasons,
but so there's
a question, you
know, between
what we actually,
you know, as in

any particular
social system,
what's actually
happening, which I
would argue could
be described
through this
technique or
what they should
be doing, which
we could also
describe through
this technique,
right?

So I think it's
important to have
that distinction
and there's a
really long history
of using dynamic
systems models to
explore those
questions, especially
in the business
context.

So that's mostly
how I've made my
living through the
decades.

So though that
that, you know,
sort of, you
know, as is to
be or would
prefer to be
sort of questioned,
it just seems
like this technique

would potentially
be a significant
advance over the
tools in that
space that currently
exists, which are
primarily things
like system
dynamics and
agent-based
modeling, stuff
like that.

But this would
seem to have a
potential to look
at things as
they are, why is
it broken, and
what we might
do?

Yes.

So I have added
several other
references.

It's important.

And the book, again,
is like, if it's
describing the
linear model, then
there's the
applications of the
linear model.

And as people, and
as John brought up
like many times in
this series of
live streams that
we did together

with four, with
just the two of
us, with John
leading through
basically the entire
thread of this
discussion of three
papers, and then two
more group
discussions, there's
been recent work
on, like, scripts
by Mao and
colleagues with a
strong and a
weak script
formalization, like
a strong script
being when the
norms are practiced
like in a
ritualistic way, like
reading from a
script, versus the
looser script
concept, which
could include just
norms and social
praxis.

And then more
recently, they're
modeling work using
PyMDP to talk
about epistemic
communities.

And then some
of the very
earliest live

streams that we
did were on
narrative, human
communication, cultural
affordances.

So these are
absolutely topics
that have been
addressed, although
many of the
earlier works are
like, here's
people in the
world, we're
going to use
active inference to
model it.

But then there's
so much more
beyond that.
it's almost like a
1950s cybernetics
book.

Well, we can just
imagine that the
person is going to
be involved in this
feedback loop.

Yeah.

But then that's not
the end of the
discussion on that.

Ali, and then

John?

I just wanted to
mention that, well,
FEP can potentially
be applied to any

dynamical system with
ergodicity.

And because you
see any system
with ergodic
behavior have
some points of
attraction, some
attractors that
all the other
states are
converging to.

So it's pretty
general and, well,
almost all dynamical
systems we know of
on different
timescales can be
potentially modeled
with FEP and as
its corollary with
active inference.

Thanks.

That's a key point,
right?

Because that's what
we look at,
ergodicity,
tough word for me.

And you can actually
look at that even in
the context of
businesses, business
strategy, and so
forth, with some
understanding of the
absorbing barriers,
right?

When does a business
dissipate?

When it's not close
enough to its
preferred state, like
adequate cash flow.

So, yeah, I think
that just seems like a
really interesting
progression from where
the state of the art
of nonlinear dynamic
systems is the way we
model businesses now.

So, well, that's the
way some of us do it.

A lot of people use a
spreadsheet.

But this would just
seem like a
potentially tremendous
advance.

Yeah.

And then just a few
notes then, John.

This Stephen Fox, who's
also joined on and
talked about industrial
engineering, he's
recently had this paper
come out just a couple
of days ago.

And ironically, a
spreadsheet is provided
in this paper, but this
is also in practice in
several places in the
world in startup

ecosystems.

systems and it's

very active

influenced.

And then one other

more conceptual note,

CAD CAD, which is a

complex systems

modeling software.

And I know some of

you are involved in

CAD CAD learning as

well.

They describe it as a

language for encoding

generalized dynamical

systems.

And that is what it

is.

Active blockference is

our project where

Jakob has done much

of the initial work

that is implementing

active inference in

CAD CAD.

And we have a whole

litany of questions

about the relationships

between Bayes graphs,

generalized dynamical

systems, variational

inference, CAD CAD,

which takes more of

like a sampling based

approach versus the

variational approaches

that we're discussing

here in active.

And it's going to be
very interesting as
this systems
engineering complex
systems simulation,
as well as tokenomic
and other things
becomes integrated
with this analytical
framework, especially
if we can demonstrate
a true concordance
between active
inference and GDS.

John?

Lyle, I just wanted
to say your comments
are really spot on,
earlier comments.

You had, I think,
if I can paraphrase,
you had essentially
said maybe even a
dysfunctional society
is performing active
inference.

And then that would
lead us to the,
really to the key
components then of
that society or that
social system is what
is its set of preferred
states and where is it
placing its attention?

So if the, if the,
if a, say, an elite

group of very wealthy people or powerful people are running the world and they're running it for their preferences, which is to maintain their wealth and to, and to ensure that, that they stay on top, well then the world, they could be performing functional active inference and the world could, you know, could, could, you know, suffer the, the, the problems that that kind of system would offer.

On the other hand, if in a highly democratic system, if people understand that they want clean air and clean water and peace and all that kind of stuff, and then that's their set of preferred states and their attention is on that and they're performing active inference.

Thank you, John.

Lyle, then Mike.

I would just throw, so John, I'm completely on board with that.

I would, I would just

amplify a little bit on
that. I would say to a
certain extent preferences
and societal decisions
are emergent.

So you, you aren't going
to find too many people
in the world who are
going to say they want
to be, you know,
starving or have war,
whatever, right?

And yet we do.

So the, the interesting
thing is how, to me,
is how does the
individual, you know,
I mean, this is the,
the whole buildup,
right?

How does the individual
behavior somehow
create an emergent
system that has many
bad outcomes?

Even though potentially
none of those are bad
actors.

Some of them are bad
actors, but let's just
take the simple case.

There are no bad actors,
yet we get bad outcome.

Why is that?

And we can observe
that in existing systems.

And then we can ask
the question, well,

you know, why is that?

And then maybe if we understand why that is, maybe we can do something to improve.

Yes.

Great point.

And then just a short note there was in our remote teams, we've framed communication and information architectures as organizational affordances.

So that could assist with decision-making, find alignment where people do have similar preferences, like a zone of proximate agreement in a negotiation context.

It might be broader, but no matter how broad it is, if you can't forage the path to it, it's going to be a failing system.

Mike?

Yeah, I just want to add, I think scaling or the scale factor is hugely important here.

So if we're talking about interactions amongst a handful of individuals, we're going to see different behaviors and outcomes

than we might see on
a global scale.
of cultural
interaction.

Okay.

And then a last
comment on that.

John, go for it.

If your hand is still
up.

I don't hear you
actually.

Okay.

Your microphone may
have changed.

Or can other people
hear, John?

Okay.

Eric, go for it.

Yeah.

How does this all
apply to a highly
distributed system?

I don't see that.

Yes.

Like a society is.

Mm-hmm.

Good question.

The multi-agent case
is something that
hasn't been studied
across the board.

but in 2021 with
colleagues, we made,
this is like one of
the early, it's not
the first multi-agent
simulation because there

were multi-agent simulations.

However, it was the first active inference model involving stigmergy, which is the feedback process with agents modifying their environments and so on.

We can think about even a nested, distributed, cognitive system as inheriting some sparsity from a causal perspective.

Like there's more connections in the Bayes graph within the nest mate's head as we're modeling it.

And then interactions induce like a sparsity between two brains and then they're always coupled with the environment and so on.

And that sparsity is a statistical convenience, but also it allows factorizability of models.

And the models may be factorized in a

way where free
energy calculations
could be, for
example, summed
within a level of
collective behavior.
I don't know if
that's going, it's
not, that's not the
solution in and of
itself to the
modeling, but when
we have systems of
like nested Markov
blankets, enclosing
systems and closing
systems, and then
also lateral
interactions and
interfaces like
collective behavior,
those systems are at
least amenable to
this kind of
principled
composition via
their sparsity,
which does map
onto a causal
architecture or at
least a useful one
of the real world.
So also remains to
be seen, but I
think that's a very
important question.
Lyle?
Eric, I would throw

in on that question
because society, of
course, broadly
speaking, is loosely
coupled.

However, of course,
there are cliques and
densely coupled areas
that are, you know,
to create the well
documented, well
documented, complex
connection graphs,
right?

So that connection
graph is emergent,
right?

So the question is, I
mean, I think this
would be actually a
really fascinating
research project.

can we show that
FEP would generate
connection graphs or
patterns of societal
interactions similar to
what has been
observed, right?

So there's lots and
lots of research on
what those graphs look
like.

So is it possible
for us to, or
someone, to use FEP
to show how those
kind of connection

patterns would
emerge?

I think that's a
really interesting
question.

Yes.

How we threshold
and parameterize
those causal graphs,
how we evaluate
counterfactuals on
those graphs, and
that's related to the
problem of structure
learning, but Bayesian
model selection and
other sort of
hierarchical Bayesian
methods have the
tools there.

Also, just quick
pointer to the
project ideas table.

There's some very
interesting projects
that people have
suggested.

Jakob, no big deal.
Sounds like a good
one.

But we can
absolutely do it,
and that's one of the
advantages of being
in, like, an
activity-driven
participatory research
and education space

projects.

We're hopefully going to be able to help each other identify those projects and where there has or hasn't been overlap, and then scaffold those projects, support them, and then provide them, like, a place to communicate about the project, playing into the ontology, active blockference, all these pieces that one researcher working by themselves would have just not those affordances.

Let's just try to quickly look at the last several questions.

So, must all preferred states and strategies be ultimately linked to survival in some way, or in the case of some organisms, can free energy be related to something other than survival?

So, just a quick

thought on that.

In the context of
repeated measurements,
measurements, yes,
survival is the
capacity for repeated
measurements.

Free energies for
domain-specific
models can be made
that have nothing to
do with survival, and
I don't think that
there's any
contradiction there.

John?

I still don't hear
you, actually.

Rohan?

And then anyone
else who raises
their hand.

Yeah, hi.

So, I was
actually wondering,
because there's no
treatment of time in
any of these methods,
right?

So, like, is there
any accommodation
made?

So, if you're
sampling something,
let's say you sample
at a minute frequency,
you're getting a
certain set of

events, whereas if you're sampling at a much higher frequency, you would get a very different, much more noisy data set, but that might be essential to capture certain things that might actually enable survival.

Maybe there's certain events that act only at certain times.

So, from the perspective of repeated measurement, right?

So, I think free energy principle, as it relates to survival, could it just be related to some sort of creating an internal clock for repeated measurement?

Great question.

Yes, so far, it's been almost time independent, as you suggest, but we're going to be dealing with two ways to deal with time in chapter four and beyond, and that's like a discrete time formalization, function, where future and past

states are explicitly modeled, partially observable, Markov decision process, and then also a continuous time variance, where future and past time points are not explicitly modeled.

However, a Taylor series approximation, higher and higher derivatives of a function are utilized to glean an approximation to the future behavior of a function.

function.

Those are two different ways of dealing with time, and then you talked about sampling rate, and I think you'll see many connections and ways to apply that.

For example, you could imagine if the sensor is every second, but you're able to think ten times per second.

Or what about when the sensor is every ten seconds, but you're only able to think over this time scale?

And it's those kinds of

situations where actually the frequencies are happening differently that we also might be able to address.

Yes, Rohan?

Yeah, yeah.

Also, like, if there are multiple sensors, some of them react quickly, some of them react slowly, or if there's just, or if we look at active inference as some sort of model averaging phenomenon, right?

So you have a model that runs really fast, that reacts really quickly, and produces predictions really quickly, and the other one acts more slowly.

Would those be valid targets in any sense?

Yes, I think that's a very interesting question.

I'll just add it here just because it was where we were discussing it.

like, about different timescales, and the way that cognitive clock rates, for example, and sensory clock rates, and then the timescale events, the statistical regularities in the world

happen.

In ModelStream 4.1, with DeVries and colleagues, they talked about the message passing in a factor graph.

So, anytime that there's a Bayes graph, there's a way to accomplish variational inference on that Bayes graph using message passing techniques.

That's what their package is about, ForneyLab.

And then, what they hinted at towards the end is actually this reactive message passing development, which uses reactive programming techniques.

So, whereas in message passing, at least in the Forney factor graph formalization, it's like you crank the model, and a message is passed amongst every node, or there's like a given execution order of messages that are passed for the entire graph.

However, the reactive message passing paradigm may allow

like, uncoupling of the frequency of message passing.

So, you might have one prior that gets one message per hour, and then there's another distribution that gets one message per minute, and another one is getting one per second, and even that one could be doing model averaging of something much faster.

So, there's many ways that I think your questions will be starting to be explored.

Rohan, and then John.

Okay, John, if your mic is back.

Can you hear me now?

Oh, good.

On the question of survival, I think you would have to identify, to answer it, you'd have to identify first, what is the organism, or what is the agent that you're talking about?

Because, obviously, if it's an individual person, a person might sacrifice their life for others, so then, you know, survival is not

the main, necessarily the main driver.

But if it's a society, then, you know, that's a, then any individual might not survive, but society as a whole might.

I suspect in the bigger picture that it's not, it's not really survival of a species.

It's probably the ability to, the propagation of information somehow.

That's what I would guess in the biggest picture.

that's probably what everything is doing is propagating information.

And Mike Levin has also explored, like, how changes in the scope of the self can lead to, like, an organism with reduced fitness as a function of a cancer with increased fitness and vision, so to speak.

And so, like, just because there's an imperative or normative way that we model behavior doesn't mean that everything works or anything like that.

Yes, Ali?

Yes, adding to what John said, well, yes, again, I believe here survival doesn't necessarily mean biological

survival.

And actually, it means, it has a much broader sense, and it means the existence of things.

actually, it's a metaphysical concept here because, as I said, Carl Friston has defined the existence or the requisites of the prerequisite of existence as having two parameters or two features, namely, being an ergodic system and having Markov blankets.

So, any system which has these, any system which has these two properties can basically be considered as an existence system or we can call it a surviving system here.

Just in our last minutes, before we go to tools, touch on some of these points.

Can Markov blanket be considered a decoupling mechanism? mechanism?

So, can anything mathematical be considered a mechanism?

People use terms in different ways.

So, just to raise the point that, like, a linear regression on height and weight isn't a mechanism of height and weight.

It's just a descriptor.

So, Markov blanket doesn't need to be any kind of mechanism at all.

It doesn't need to cut nature at

the joints or anything like that.

That being said, there are ways in which one could think of the blanket, again, variously, depending on how the real generative model and generative process separated as part of a Markov partition, sometimes there's ways to think of it as how they're coupled.

Other times, it is a statistical decoupling of a type.

It's a partial information encapsulation, but it's not a total encapsulation.

And that way, it is like an interface.

And yes, thanks to whomever has added this quote, this is one reason why the transcripts of all of the live streams and discussions augmented with the ontology will be so helpful and searchable because we could find sentences that have raised similar points and address them.

So, good question.

And then just last ones.

Maybe we can come back to this.

And then this one is a nice next step question.

Thinking about friction, what is the analogy to friction in a cognitive system?

What is the stochastic fluctuation on our cognitive path of least action?
what is that in the real world?

Realism, ontological claims?

And then from the behavioral modeling perspective, where are we using these equations and partitioning variance into a path or a flow of least action and a variable that is analogous to stochastic fluctuation or friction, but doesn't take on any of the baggage of describing some feature or component or mechanism of the physical system?

So, also a great question though.

And we'll leave this one in case people want to continue to address it.

So, that brings us to the end of chapter three, live meetings.

Next week, we'll head into two weeks of chapter four, then two weeks of chapter five, and then we'll have two weeks to sort of ground ourselves, think about what we've learned, talk about project ideas, and then we'll be heading into the second cohort of the first half of the book.

And for those who are here who want to continue into the back nine, as it were, the second half of the textbook.

And also people are welcomed and encouraged to be facilitators or return participants in the second cohort first part, because it's many coats of paint for everyone.

And so, we want to provide that affordance to return again and again and again, because the basics are the classics and the essence and the seed of innovation, all this other important things that happen when we return and clarify.

Also, we've had to sort of do all of these entering of figures and equations for the first time.

But now that we have the architecture, people can

focus on annotating them, or they can focus on adding new questions or improving the questions that others have written.

If anyone else has any comments that they want to make while the recording is still occurring, I'll just give a few seconds for anyone to raise their hand. And then we'll end this session and then take a one minute break.

And then in this room, we'll be continuing on with tools organizational unit.

And if you want to continue speaking with people here about just any topic at all, you can head up to a room up top.

And then, of course, always you can join into the Discord voice channels and just be working on things like during the math reading group or any other area.

Rohan, and then anyone else who raises their hand.

Yeah.

So, I actually, so I'm sorry for this late thought.

So, I'm just curious.

So, when you say the free energy principle, it's basically figuring out some set point of states, internal states that the agent wants to be with, right?

It's not necessary that it actually couples to the real world.

So, why wouldn't it be that, okay, it just becomes a brain in the vat type of situation.

So, it just needs to find a set point that helps it survive.

So, it doesn't have to actually model the real world.

It just has to find a place and no matter what happens, just stay there.

So, it's not really dynamic.

So, that could be, is that a possibility?

It is dynamic.

It is explicitly modeling time.

And one can make a model just like a linear regression could be static or you could use the linear regression technique and use it in

a time-series analysis.

No, I was more referring it.

So, if you look at some of the reinforcement, deep reinforcement learning models, right?

They usually find some trick within the game.

Like, if you're looking at Atari games and you say maximize the score, for example.

So, it tries to find, you know, okay, I can maximize the score by doing the simplest thing that doesn't require me to progress in any way.

But it would increase the score.

Okay.

So, yes, these are important things.

So, write them down so that we can have infinite discourse on them.

The brain of that does still have an interface with the environment, whether it's receive only.

And then you brought up, like, it doesn't have to be, like, modeling the, quote, real world.

This is the architecture that helps us bypass that and say, right, the particulate states, the particular states, which are the blanket states and the internal states, are doing inference, acting as if they're doing inference, on external states.

So, it's perfectly compatible with scientific realism, scientific surrealism, aesthetic surrealism.

You could, but evolution and just a bit of systems will sweep it off the table.

So, I mean, the states could be hallucinating.

It could be hallucinating the world and it's actually just doing inference on the hallucination rather than actual reality as such.

Anil Seth and many others who speak of that reality as a controlled hallucination.

They're two feet in that territory without any qualms and feeling like it addresses questions that have been unaddressed by other perspectives.

Okay.

Okay.

Okay.

Yeah.

Okay.

So, I'll just put this down.

Yeah.

Cool.

Thank you, everybody.